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EVALUATING J85 ENGINE THRUST CHART ACCURACY FROM FLIGHT TEST DATA

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EVALUATING J85 ENGINE THRUST CHART ACCURACY FROM FLIGHT TEST DATA

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To my beloved wife and my precious daughter, you both are my greatest inspiration.

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"“Success is not final, failure is not fatal:
it is the courage to continue that counts."
— Winston Churchill

Resumo

A precisão das estimativas de desempenho em voo é crucial para o planejamento eficaz de missões e a segurança operacional da aeronave F-5. Observações de voos experimentais indicaram discrepâncias nos dados de subida do motor J85-GE-21, especialmente no regime militar (MIL), o que motivou uma investigação detalhada sobre essas diferenças. Este estudo tem como objetivo avaliar a exatidão das previsões do manual de desempenho, comparando-as com dados coletados em voos reais. A análise focou-se em dois regimes de operação: MIL e Pós-Combustão (PC). No regime MIL, verificou-se que os tempos de subida, consumo de combustível e distâncias reais superaram consistentemente os valores estimados pelo manual. Já no regime PC, os dados experimentais apresentaram menor discrepância com as estimativas do manual, indicando maior precisão. Foram analisados os perfis de velocidade, e o empuxo líquido foi estimado pelo método Mass Consumption Acceleration (MCA), fundamentado na conservação da energia específica. O modelo paramétrico desenvolvido para representar o empuxo demonstrou alta precisão, alcançando um coeficiente de determinação de 0,98 no regime MIL e de 0,95 no regime PC, fornecendo estimativas confiáveis em ambos os regimes durante a subida. A comparação com os dados de empuxo bruto do deck do motor revelou uma diferença média de 22% no regime MIL e de 8% no regime PC. Embora diferenças entre empuxo bruto e líquido sejam esperadas, a magnitude da discrepância no regime MIL sugere que o motor opera abaixo do desempenho previsto.

Abstract

The accuracy of in-flight performance estimates is crucial for effective mission planning and operational safety of the F-5 aircraft. Observations from experimental flights indicated discrepancies in the climb data for the J85-GE-21 engine, especially in the Military Thrust (MIL) regime, prompting a detailed investigation into these differences. This study aims to assess the accuracy of the performance manual's predictions by comparing them with data collected from real flights. The analysis focused on two operational regimes: MIL and Afterburner (AB). In the MIL regime, it was found that the actual climb times, fuel consumption, and distances consistently exceeded the estimated values provided in the manual. In the AB regime, however, the experimental data showed less discrepancy with the manual's estimates, indicating higher accuracy. The speed profiles were analyzed, and the net thrust was estimated using the Mass Consumption Acceleration (MCA) method, based on the conservation of specific energy. The parametric model developed to represent the thrust demonstrated high accuracy, achieving a coefficient of determination of 0.98 in the MIL regime and 0.95 in the AB regime, providing reliable estimates for both regimes during climbs. Comparison with the gross thrust data from the engine deck revealed an average difference of 22% in the MIL regime and 8% in the AB regime. Although differences between gross and net thrust are expected, the magnitude of the discrepancy in the MIL regime suggests that the engine operates below the expected performance level.

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List of Abbreviations and Acronyms

AB	Afterburner.
EFMA	Excess Force Mass Acceleration.
EGT	Exhaust gas temperature.
FF	Fuel flow.
IPEV	Instituto de Pesquisas e Ensaios em Voo.
ISA	International Standard Atmosphere.
MCA	Mass Consumption Acceleration.
MIL	Military Thrust.
RPM	Revolutions per minute.
TAS	True Airspeed.
TOT	Total Fuel.

List of Symbols

a	Acceleration.
A	Nozzle area.
α	Angle of attack.
A_R	Aspect Ratio.
C_D	Drag coefficient.
C_{D_0}	Parasite drag coefficient.
C_L	Lift coefficient.
D_A	Interference drag from the exhaust nozzle.
δ	Pressure ratio.
Δ	Nozzle opening.
D_I	Inlet drag.
D	Aerodynamic drag force acting against the aircraft's movement.
D_{ram}	Ram drag.
e	Oswald efficiency factor.
$E_{\mathbf{mec}}$	Mechanical energy.
η_b	Burner efficiency.
η_c	Compressor efficiency.
η_l	Intake efficiency.
η_m	Mechanical efficiency.
η_t	Turbine efficiency.
$F_{\mathbf{excess}}$	Excess force.
$F_{\mathbf{tan}}$	Tangential force along the aircraft's flight path.

γ	Flight path angle.
H	Specific energy.
L	Lift force.
m	Mass.
M_a	Mach number.
μ	Air viscosity.
N_r	Engine rotational speed.
N_z	Load factor.
P_0	Reference pressure.
P_t	Ambient pressure.
R	Specific gas constant.
ρ	Air density.
T_G	Gross thrust.
TG_{std}	Standard gross thrust.
θ	Temperature ratio.
T_n	Net thrust.
T_t	Ambient temperature.
V_c	Calibrated airspeed.
V_f	Final velocity.
V_i	Initial velocity.
V_t	True airspeed.

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1 Introduction

The Instituto de Pesquisas e Ensaios em Voo (IPEV) is a Brazilian military organization whose main activity is inflight aircraft testing. The IPEV is responsible for certifying newly manufactured fixed- or rotary-wing aircraft by verifying the operation of its subsystems in flight conditions and testing the integration of new features, weapons, or critical technologies. The Flight Test Course, conducted by the Brazilian Air Force's IPEV, aims to train qualified pilots and engineers to plan, execute, and manage flight testing activities for the development, modification, evaluation, or certification of aircraft and onboard systems. One of these aircraft is the F-5M, which has been extensively used in flight test campaigns. During these campaigns, historical discrepancies have emerged in the calculations for climb performance in the MIL regime. The MIL regime represents the engine's maximum thrust output without engaging the afterburner, delivering high power with fuel efficiency. These discrepancies appear as consistent underestimations of climb time, distance, and fuel consumption when compared to actual flight data.

1.1 Motivation

The importance of having a reliable performance estimation tool that accurately reflects the real-world data of the aircraft cannot be overstated. Accurate performance predictions are essential for effective mission planning, especially in conflict scenarios where missions must often adhere to strict timelines. Any deviation from the planned schedule can compromise mission success and potentially endanger both the aircraft and crew. Inaccurate performance estimations have several critical implications. They can lead to suboptimal mission planning, resulting in insufficient fuel reserves, miscalculated time-to target, and incorrect assessments of aircraft capabilities. This is particularly problematic for the F-5, which has a limited range and often operates at the edge of its fuel capacity. Moreover, these discrepancies pose a significant risk to flight safety. The F-5's restricted autonomy means that any miscalculation of fuel consumption or climb performance could lead to situations where the aircraft flight with low fuel before completing its mission or returning to base. This risk is exacerbated during military operations where deviations

from the planned flight profile can result in unforeseen operational hazards. Thus, addressing these discrepancies is not just a matter of improving computational models but is also a critical aspect of ensuring mission success and flight safety.

1.2 Objectives

The objective of this work is to investigate the discrepancies between the flight manual data and experimental flight test results for climb performance in the MIL regime of the F-5 fighter aircraft equipped with J85-GE-21 engine. This study will identify and understand the causes of these discrepancies, with the goal of improving the accuracy of performance estimates and ensuring that they align more closely with real-world data. By thoroughly analyzing these discrepancies, this research seeks to enhance mission planning and flight safety for the F-5.

1.3 Work Organization

This work is organized into five chapters, as follows:

- Chapter 1 - Introduction: Provides an overview of the motivation, objectives, and structure of the work.
- Chapter 2 - Literature Review: Covers the relevant background information on the F-5 aircraft, the J85-GE-21 engine, performance manual data, and related studies.
- Chapter 3 - Methods: Explains the methodologies used to estimate time, fuel, and distance, along with the thrust estimation approach, including longitudinal flight mechanics, parametric models, and parameter estimation methods.
- Chapter 4 - Results and Discussion: Presents the findings of the study, comparing experimental results with flight manual data and analyzing discrepancies.
- Chapter 5 - Conclusions: Summarizes the main findings, discusses their implications, and provides recommendations for future research.

2 Literature Review

2.1 The F-5 Aircraft

The F-5M is a high-performance, multipurpose tactical fighter aircraft, initially produced by Northrop Corporation and later modernized by Embraer S.A. This aircraft is designed to perform air-to-air and air-to-ground missions. It is equipped with advanced avionics and weapon systems.



FIGURE 2.1 – The F-5M Aircraft (FAB, 2024)

The F-5M is equipped with fuselage fuel cells and additional external tanks, allowing for extended operational range. The aircraft is also fitted with a probe for air refueling, which significantly increases its endurance and flexibility during missions. The standard internal fuel capacity is approximately 2,983 pounds(1,353 kg), and with external tanks, it can carry up to 4,424 pounds(2,007 kg) of fuel, extending its operational range significantly.

The F-5M's armament includes a 20mm cannon in the nose of the aircraft and air-to-air missiles in the wingtip stations and external pylons. The aircraft is also capable of carrying bombs, rockets and flares on its five jettisonable pylons below the wings.

In terms of performance, the F-5M can reach a maximum speed of Mach 1.6 and it's service ceiling is around 51,800 feet (15,800 meters). The F-5M exists in two versions: a single-seat version and a dual-seat version, known as F-5FM. The dual-seat version is primarily used for training purposes but retains full combat capabilities, allowing for seamless transition from training to operational missions (EMBRAER, 2010; FAB, 2024).

TABLE 2.1 – Aircraft dimensional and performance data (EMBRAER, 2010; FAB, 2024)

Dimensions		
Length	14.68 m	48 ft 2 in
Wingspan	8.13 m	26 ft 8 in
Height	4.06 m	13 ft 4 in
Wing area	18.062 m^2	194.40 ft^2
Gross weight	3,275 kg	15,549 lb
Performance		
Maximum speed	Mach 1.63	
Propulsion	20.7 kN	4,650 lbf
Fuel capacity	2,642 l	698 gal
Service ceiling	15,800 m	51,800 ft

2.2 The J85-GE-21 Engine

The J85-GE-21 engine, used in the F-5 aircraft, is a turbojet engine equipped with afterburners. It delivers a static thrust of 3250 pounds(14.4 kN) at military power and up to 4650 pounds(20.7 kN) at maximum afterburner power under sea level, standard day conditions. The engine features a nine-stage axial flow compressor, directly coupled to a two-stage turbine. The air intake for each engine is facilitated through ducts on the side of the fuselage, directing air into the compressor section using a variable geometry system comprising inlet guide vanes and variable stator vanes. This system helps to reduce the likelihood of a compressor stall.

The throttle system of the J85-GE-21 engine provides main engine control from OFF to IDLE, IDLE to MIL, and from minimum to maximum afterburner operation. Each throttle controls the respective engine's fuel supply, fuel shutoff valve, main fuel control throttle angle, stopcock valve, main and afterburner ignition circuitry, engine speed, and afterburner control operation.

The J85-GE-21 engine employs an annular combustion chamber, which includes 12 fuel injectors that spray liquid fuel into the chamber at high pressure The two-stage turbine extracts energy from the high-temperature, high-pressure gases to drive the compressor

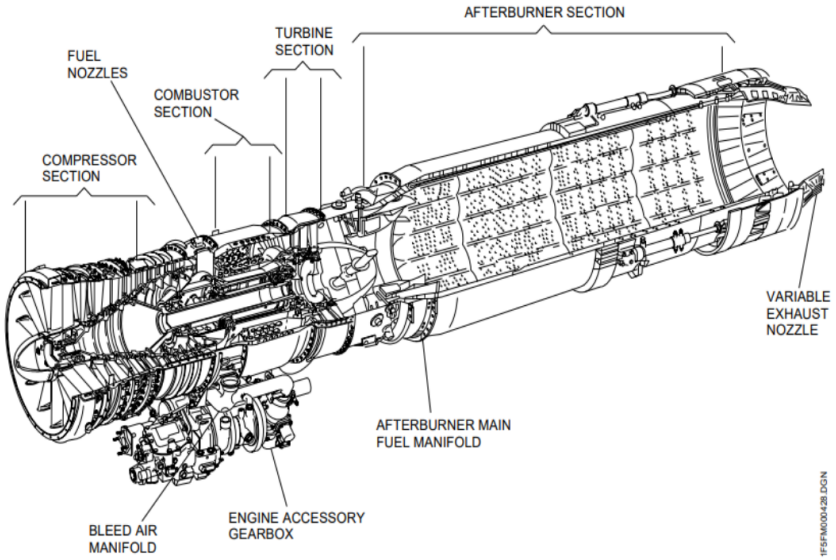


FIGURE 2.2 – The J85-GE-21 Engine (EMBRAER, 2010)

and other accessories such as hydraulic pump and generator. Additionally, 20 fuel injectors are used in the afterburner section to provide additional thrust. Exhaust gases from the combustor section pass through the turbine section and are expelled via a variable exhaust nozzle. The EGT control system adjusts the nozzle opening electrohydraulically to prevent overtemperature and maintain EGT within safe limits during MIL and afterburner power operations.

To mitigate the risk of compressor stalls, the J85-GE-21 engine incorporates variable inlet guide vanes and variable stators, which automatically adjust based on engine RPM and inlet temperature. Additionally, a P3 compressor dump system activates during throttle bursts to the afterburner range at intermediate or high altitudes, further reducing the possibility of stalls (EMBRAER, 2010).

2.3 Performance Manual

The aircraft's performance manual provides detailed guidance and essential parameters for assessing performance across different flight phases, including takeoff, climb, cruise, and landing. Within this manual, the climb charts offer critical data for calculating performance under varying drag indexes and aircraft weights. These charts are used to determine time, distance, and fuel consumption during a climb and apply to both maximum and military thrust conditions, ensuring accurate and reliable performance calculations throughout flight operations.

The drag index system is integral for calculating aircraft performance, as it quantifies the aerodynamic drag caused by external loads and configurations. Each component,

such as wingtip stores, centerline stores, or external fuel tanks, is assigned a specific drag number based on its size, shape, and position on the aircraft. These drag numbers are summed to calculate the total drag index, which directly affects the aircraft's fuel consumption, range, and climb rate.

These climb charts present performance data for the aircraft from sea level to high altitudes, accounting for a range of gross weights and drag indexes from 0 to 400. They include schedules for maximum thrust climbs, aiming for minimum time ascent, as well as military thrust climbs, which prioritize minimal fuel consumption. Additionally, the charts cover single-engine maximum thrust climbs for drag indexes ranging from 0 to 120, focusing on maximizing the aircraft's range (EMBRAER, 2010).

For both maximum and military thrust conditions, the manual provides distinct charts for each scenario. The data is organized into three key parameters—fuel used, time to climb, and distance traveled—based on sea level gross weight, pressure altitude, drag index, and temperature. Temperature correction scales ensure accurate results for non-standard day conditions, adjusting performance metrics based on environmental variations.

As the aircraft's weight or drag index increases, time to climb, fuel consumption, and distance also rise due to the greater energy required to overcome increased drag and weight. Additionally, higher than International Standard Atmosphere (ISA) temperature decrease engine performance and air density, further extending climb times and fuel consumption.

When starting to climb from an altitude other than sea level, the fuel required is determined by calculating the difference in fuel needed to climb from sea level to the target altitude. The values for fuel, time, and distance are adjusted for sea level gross weight, incorporating the additional fuel necessary to climb from sea level to the initial starting altitude.

To calculate climb performance under specific conditions, users follow a systematic approach: entering the charts with the required parameters, applying temperature corrections as needed, and interpolating values for more precise results.

Figure 2.3 illustrates the process for calculating climb performance. Starting with the sea level gross weight, the process moves through pressure altitude, drag index, and Delta ISA, concluding with the determination of the fuel, distance, and time required for the climb.

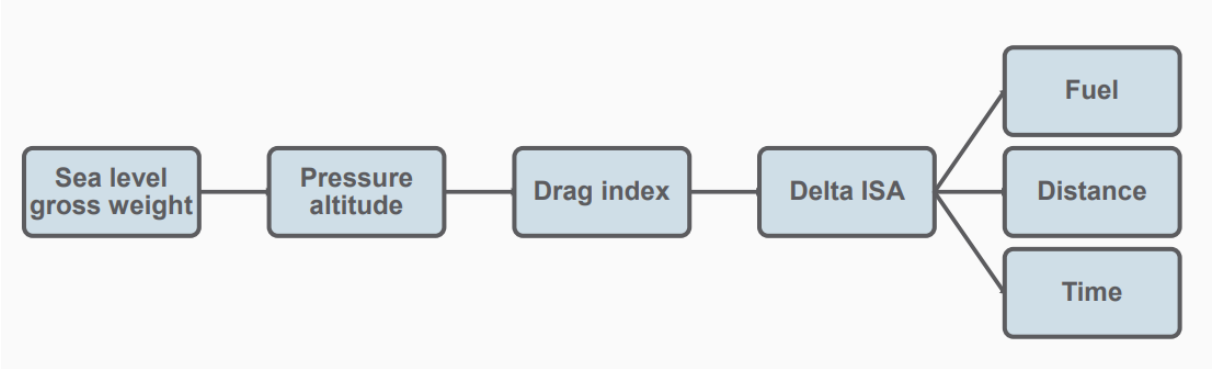


FIGURE 2.3 – Diagram representing the steps for calculating climb performance.

2.4 Inflight thrust measurement technics

Estimating in-flight thrust is crucial for evaluating the aircraft's performance, especially when determining parameters like fuel efficiency, climb rate, and aerodynamic characteristics. Two key methods for performing this estimation are the MCA method and the EFMA method. These methods differ in their underlying principles and approaches to calculating thrust, with each offering specific advantages depending on the available data and flight conditions.

2.4.1 The MCA method

The MCA method is based on the principle of energy conservation, focusing on the conversion of thermal energy into mechanical energy. In this method, thrust is estimated by analyzing the variation in the aircraft's mechanical energy using data on acceleration, drag, and fuel consumption. The core equation relating mechanical energy E_{mec} to thrust is:

$$\Delta E_{\text{mec}} = T_n \cdot \Delta x - D \cdot \Delta x$$

Here, T_n is the thrust force to be estimated, and D is the aerodynamic drag force. The change in mechanical energy ΔE_{mec} is calculated from the variation in the aircraft's velocity and altitude:

$$\Delta E_{\text{mec}} = \frac{1}{2} m \cdot ({V_f}^2 - V_i^2)$$

where:

- V_f is the final velocity of the aircraft,

- V_i is the initial velocity of the aircraft, and
- m is the mass of the aircraft.

The thrust is inferred based on the energy converted into acceleration. MCA is effective in situations where engine models are unavailable, and its simplicity makes it useful for a wide range of applications. However, it relies on assumptions such as constant aircraft mass and steady aerodynamic conditions, which can limit its accuracy in dynamic or complex flight scenarios (Rosenberg; Schucht, 1985)

2.4.2 The EFMA Method

The EFMA method provides a more direct approach to estimating thrust. Instead of relying on energy conversion, EFMA calculates the tangential force acting along the flight path, which includes the thrust component. This tangential force is the sum of the aerodynamic drag force and the excess force generated by the difference between engine thrust and air resistance. The basic EFMA equation is:

$$F_{\text{tan}} = D + F_{\text{excess}}$$

In this case, F_{tan} represents the tangential force along the flight path and F_{excess} is the additional force measured directly from the aircraft's acceleration:

$$F_{\text{excess}} = m \cdot a$$

With this, EFMA directly calculates thrust based on acceleration and flight dynamics data. One of the main advantages of EFMA is that it does not rely on pre-existing engine or airframe models, allowing thrust to be calculated using only flight data collected during specific maneuvers, such as accelerations, decelerations, and dynamic maneuvers (e.g., banked turns or split-S maneuvers) (Schuch; Weiss; Krause; Thurling, 2001; Vitsas, 2015).

3 Methods

3.1 Estimation of Time, Fuel Consumption, and Distance

The estimation of time, distance, and fuel consumption was conducted using data from the acquisition system, which operates at a frequency of 32 Hz. To ensure the reliability and accuracy of the data, a series of pre-processing steps were implemented. Initially, adjustments were made to transform the reference frame from the global system to the body-fixed coordinate system of the aircraft. Additionally, appropriate unit conversions were applied to standardize all parameters, facilitating accurate calculations and comparisons.

Key parameters were calculated to support the estimation process. The air distance was obtained by integrating the True Airspeed (TAS) over time, providing an estimate of the distance traveled through the air. True Airspeed represents the actual speed of the aircraft relative to the surrounding air, taking into account variations in air density due to altitude and temperature. In parallel, the ground distance was calculated by integrating the Ground Speed, which reflects the aircraft's actual speed over the earth's surface, including the effects of wind. The Total Fuel (TOT) parameter was determined by integrating the Fuel flow (FF) from both engines.

In addition to these parameters, other engine performance data were derived from the engine deck provided by the manufacturer. Engine decks, also known as engine performance decks, are tables or charts that detail the engine's behavior under various operating conditions. These data sets are essential for engineers and aircraft operators, as they assist in mission planning and performance analysis during different flight phases.

Through input data such as Mach number, altitude, engine Revolutions per minute (RPM), outside air temperature, and nozzle area, key performance outputs can be derived, including thrust, fuel flow, and exhaust gas temperature. In cases where segments of the engine deck data were incomplete or invalid, linear interpolation techniques were applied to ensure continuity and reliability. The engine deck is particularly useful in flight tests, allowing measured results to be compared with the manufacturer's predictions.

Once the data had been thoroughly pre-processed and cleaned, the climb time was calculated by determining the difference in elapsed time between the selected starting and ending altitudes. Similarly, the total fuel consumption was determined by calculating the difference in the TOT on board between these two altitudes. The distance covered during the climb was obtained by calculating the difference in the ground distance parameter, which represents the actual distance traveled over the terrain.

Table 3.1 presents a summary of the data provided by the F-5 aircraft's acquisition system, with each variable described in terms of its unit, associated symbol. The variables listed in the table cover essential aspects of the aircraft's performance, including speed, altitude, orientation, atmospheric conditions, and engine parameters.

TABLE 3.1 – Data Table Acquired by Internal System

Variable	**Name**	**Unit**	**Description**
01	UTC	s	Coordinated Universal Time
02	Zpb	ft	Barometric pressure altitude
03	Vc	KCAS	Calibrated airspeed
04	MACH	-	Mach number
05	BAROSET	inHg	Barometric setting
06	TAS	KTAS	True airspeed
07	ALFA_TRUE	deg	True angle of attack
08	WIND_DIRECTION	deg	Wind direction
09	WIND_SPEED	kt	Wind speed
10	TEMP_STATIC	°C	Static temperature
11	PRESS_STATIC	Pa	Static pressure
12	PRESS_TOTAL	Pa	Total pressure
13	ALFA_IND	deg	Indicated angle of attack
14	TI	C	Indicated temperature
15	GROUND_SPEED	kt	Ground speed
16	RALT	ft	Radar altitude
17	PHI	deg	Inclination angle
18	PITCH	deg	Pitch angle
19	PSI	deg	Yaw angle
20	P	deg/s	Roll rate
21	Q	deg/s	Pitch rate
22	R	deg/s	Yaw rate
23	LAT	deg	Latitude
24	LONG	deg	Longitude
25	NORMAL_ACCEL	g	Normal acceleration
26	LFF	lb/h	Left engine fuel flow

Variable	Name	Unit	Description
27	RFF	lb/h	Right engine fuel flow
28	LFQT	lb	Left fuel tank quantity
29	RFQT	lb	Right fuel tank quantity
30	EGT_E	°C	Exhaust gas temperature left engine
31	EGT_D	°C	Exhaust gas temperature right engine
32	RPM_E	%	Left engine RPM
33	RPM_D	%	Right engine RPM
34	POLEO_E	psi	Left engine oil pressure
35	POLEO_D	psi	Right engine oil pressure
36	NOZZLE_E	%	Left engine nozzle area
37	NOZZLE_D	%	Right engine nozzle area
38	NX	g	Longitudinal acceleration (earth reference)
39	NY	g	Lateral acceleration (earth reference)
40	NZ	g	Vertical acceleration (earth reference)
41	MAG_HEAD	deg	Magnetic heading
42	GND_TRACK	deg	Ground track
43	nx_body	g	Longitudinal acceleration (body reference)
44	ny_body	g	Lateral acceleration (body reference)
45	nz_body	g	Vertical acceleration (body reference)
46	Weight	lb	Aircraft total weight
47	CG	%	Center of gravity
48	Dist AR	NM	Air distance
49	Dist GND	NM	Ground distance
50	TOT	lb	Total fuel
51	J85_Thrust_E	lb	Thrust of left engine (engine deck)
52	J85_Thrust_D	lb	Thrust of right engine (engine deck)
53	J85_Fuel_flow_E	lb/h	Left engine fuel flow (engine deck)
54	J85_Fuel_flow_D	lb/h	Right engine fuel flow (engine deck)
55	J85_EGT_E	°C	Left engine exhaust gas temperature (engine deck)
56	J85_EGT_D	°C	Right engine exhaust gas temperature (engine deck)
57	THRUST	lb	Total thrust (engine deck)
58	ALFA_CORR_DEG	deg	Corrected angle of attack for MACH number
59	TaISA	C	Standard atmosphere temperature
60	nxa	g	Longitudinal acceleration (aerodynamic reference)
61	nza	g	Vertical acceleration (aerodynamic reference)
63	CL	-	Lift coefficient
64	CD	-	Drag coefficient

3.2 Thrust Estimation Methodology

To evaluate the various parameters that might affect climb performance, a methodology was proposed to estimate the thrust generated by the propulsion system. This information allows for a comparative analysis with the thrust calculated by the engine's performance deck, which could be useful in identifying potential discrepancies between the expected engine thrust and the actual performance, possibly contributing to the observed climb data discrepancies.

Direct measurement of aerodynamic and propulsive forces during flight is not feasible. To determine the magnitude and effects of these forces, they must be modeled and calculated based on flight data. The application of mechanical energy analysis, along with thrust and drag models, allows for the estimation of these forces.

3.2.1 Longitudinal Flight Mechanics

In a climbing aircraft with a nearly constant speed, the main forces acting on the aircraft are the gravitational force, the aerodynamic forces, and the propulsive force. The aerodynamic forces can be decomposed in the aerodynamic reference frame, where lift acts perpendicular to the wings, and drag opposes the aircraft's motion along the x_a axis of the aerodynamic reference frame.

During steady climbing flight, the balance of forces includes the aircraft's weight mg, acting downward, the lift L, which supports the aircraft's weight, and the drag D, which opposes its forward motion. Thrust T_n is required to overcome both drag and the component of weight acting in the direction of the climb, represented by the flight path angle γ.

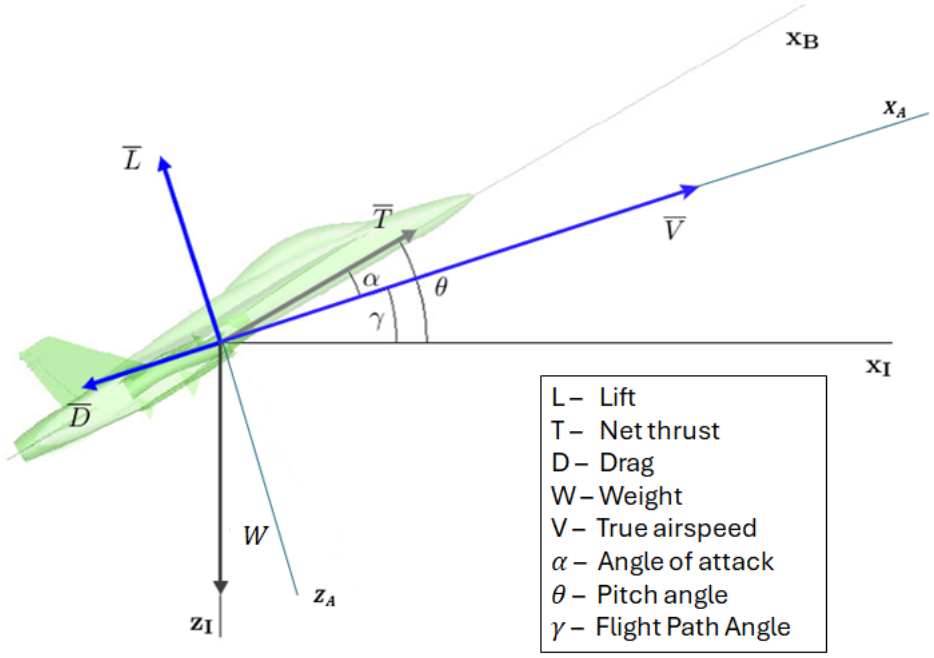


FIGURE 3.1 – Free body diagram of an aircraft in steady climbing flight (McClamroch, 2006)

Decomposing the forces in the aerodynamic coordinate system yields the following equations:

$$\sum F_{x,\mathrm{a}} = T_n \cos(\alpha) - D - W \sin(\gamma) = \frac{W}{g} \cdot \dot{V}_t \tag{3.1}$$

$$\sum F_{z,\mathrm{a}} = W \cos(\gamma) - L - T_n \sin(\alpha) = \frac{W}{g} \cdot V_t \cdot \dot{\gamma} \tag{3.2}$$

To better analyze the differentiation between propulsive force and drag acting along the same axis, the MCA will be employed. This method is based on the principle of energy conservation, which states that the change in a body's mechanical energy is equivalent to the work done by external forces. This allows to decompose the forces acting on the aircraft and compute the thrust required for a given flight condition (Rosenberg; Schucht, 1985; Vitsas, 2015).

The work done by an external force is expressed as the integral of the force applied over the distance through which it acts:

$$W = \int F_{ext} \cdot dx = dE_{\mathrm{mec}} \Rightarrow F_{ext} = \frac{dE_{\mathrm{mec}}}{dx} = \frac{dE_{\mathrm{mec}}}{dt}\frac{dt}{dx}$$
$$\Rightarrow \frac{dE_{\mathrm{mec}}}{dt} = F_{ext}\frac{dx}{dt} \tag{3.3}$$

In this case, F_{ext} represents the external forces acting on the aircraft, including thrust and drag as described in the Equation 3.1.

$$\frac{dE_{\text{mec}}}{dt} = (T_n \cos(\alpha) - D - W \sin(\gamma)) \cdot \frac{dx}{dt}$$

The mechanical energy of the aircraft, in terms of its velocity, is given by the equation:

$$E_{\text{mec}} = m\frac{{V_t}^2}{2}$$

Taking the derivative of this expression with respect to time:

$$\frac{dE_{\text{mec}}}{dt} = mV_t\frac{dV_t}{dt} \tag{3.4}$$

The MCA method heavily relies on this relationship, utilizing the idea that the change in mechanical energy is driven by the net force acting on the aircraft, which includes thrust along the longitudinal axis and drag that opposes motion (Rosenberg; Schucht, 1985). By substituting $\frac{dE_{\text{mec}}}{dt}$ and using the true airspeed $V_t = \frac{dx}{dt}$, it's obtained:

$$mV_t\frac{dV_t}{dt} = (T_n \cos(\alpha) - D - W \sin(\gamma)) \cdot V_t$$

$$W\left(\frac{V_t}{g}\frac{dV_t}{dt} + \sin(\gamma) \cdot V_t\right) = (T_n \cos(\alpha) - D) \cdot V_t$$

$$\left(\frac{V_t}{g}\frac{dV_t}{dt} + \sin(\gamma) \cdot V_t\right) = \frac{(T_n \cos(\alpha) - D)}{W} \cdot V_t$$

Replacing $\sin(\gamma) \cdot V_t$ with $\frac{dh}{dt}$ gives:

$$\left(\frac{V_t}{g}\frac{dV_t}{dt} + \frac{dh}{dt}\right) = \frac{(T_n \cos(\alpha) - D)}{W} \cdot V_t$$

This equation simplifies to reveal the specific components of energy change due to thrust and drag over time. In addition, the rate of change in specific energy, which combines changes in kinetic and potential energy, can be expressed with the following equation:

$$\dot{H} = \frac{V_t}{g}\frac{dV_t}{dt} + \frac{dh}{dt} \tag{3.5}$$

This equation represents the rate of change in the aircraft's specific energy, where the first term $\frac{V_t}{g}\frac{dV_t}{dt}$ describes the rate of change in kinetic energy as a function of acceleration and velocity, and the second term $\frac{dh}{dt}$ represents the change in potential energy due to

altitude.

By rearranging the equation, the following relation is obtained:

$$\frac{\dot{H}}{V_t} = \left(\frac{T_n \cos(\alpha) - D}{W} \right) \tag{3.6}$$

This equation allows the calculation of the rate of change in the aircraft's specific energy at any moment during flight, based on the propulsive and resistive forces relative to the aircraft's weight. By knowing the varying values of speed, altitude, and mass over time, the aircraft's energy behavior can be calculated accurately (Rosenberg; Schucht, 1985; Vitsas, 2015).

3.2.2 Parametric Model of Thrust

For thrust modeling, it is essential to choose a model that can represent the complex nonlinear behavior resulting from the variation of individual parameters. A key requirement for any engine thrust model is that it must exhibit smooth behavior without significant local curvature changes. Additionally, the model must account for the environmental factors that significantly influence thrust output, such as pressure, temperature, and velocity.

Ambient pressure affects thrust by altering the air density entering the engine. As altitude increases and pressure decreases, the lower air density reduces the mass flow rate, thus decreasing thrust. Temperature also impacts air density; higher temperatures reduce air density, lowering oxygen availability for combustion and, consequently, reducing thrust. Velocity, on the other hand, influences the intake of air due to ram pressure. At higher velocities, the increased air pressure at the intake can enhance thrust up to a point, but at supersonic speeds, the associated drag begins to counteract these benefits. These environmental factors are encapsulated in parameters such as the pressure ratio (δ) and the temperature ratio (θ), which are critical for accurate thrust modeling (Saravanamuttoo; Rogers; Cohen, 2001).

The general expression for thrust, accounting for these variables and others specific to the engine configuration, can be written as (USAF, 1993):

$$T_n = f(V_t, T_t, P_t, N_r, A, R, \mu, \eta_l, \eta_c, \eta_b, \eta_t, \eta_m) \tag{3.7}$$

Since the component efficiencies are primarily functions of V_t, T_t, P_t, N_r, and A, and because μ is mainly a function of T_t and P_t,he equation above can be simplified to express thrust as a function of its primary variables:

$$T_n = f(V_t, T_t, P_t, N_r, A, R) \tag{3.8}$$

The functional relationship described by Equation 3.8 enables the development of parametric models that can estimate thrust with reasonable accuracy under most flight conditions. To account for nonlinear correlations, a general polynomial mathematical model, as shown in Equation 3.9, was used. It is expected that the polynomial model will be able to estimate engine thrust more accurately under different flight conditions, including those where the relationship between thrust and the input variables is highly nonlinear. This model, although more complex, offers a more faithful representation of reality. Furthermore, using a polynomial model better accommodates the variability observed in the data, providing a robust and reliable thrust estimate throughout the flight envelope. It is important to note, however, that the model's complexity must be balanced with practicality and interpretability, making careful data analysis and system behavior evaluation under different operational conditions essential (Domingos, 2023).

$$T_n = T_0 \left(\sum C_{NJK} \cdot \left(\frac{N_r}{\sqrt{\theta}} \right)^N \cdot {M_a}^J \cdot \Delta^K \right) \tag{3.9}$$

For this work, a 3rd degree polynomial parametric model was proposed as described in the equation 3.10.

$$\frac{T_n}{\delta} = T_0 \cdot \left[C_0 + \sum_{i=1}^{3} \left(C_{i,1} \cdot \left(\frac{N_r}{\sqrt{\theta}} \right)^i + C_{i,2} \cdot {M_a}^i + C_{i,3} \cdot \Delta^i \right) \right] \tag{3.10}$$

where:

- T_n is the net thrust produced by the engine,
- δ represents the pressure ratio relative to standard atmospheric pressure,
- T_0 is the reference thrust at sea level and standard conditions,
- C_0 and $C_{i,j}$ are empirical coefficients determined through regression analysis,
- N_r is the engine rotational speed,
- θ denotes the temperature ratio relative to standard atmospheric temperature,
- M_a is the Mach number,
- Δ is the nozzle opening.

3.2.3 Parametric Model of Drag

Aerodynamic drag refers to the resistive force that an object in motion encounters when interacting with the surrounding fluid, such as air. It is a force that acts in the opposite direction of the motion of the object, limiting its speed and influencing its performance. The origin of drag is associated with two main phenomena:

- **Pressure Drag:** This component of drag is generated by the pressure differential along the moving body. Due to the static pressure difference between the front and rear of the body, a resultant force opposes the motion.
- **Friction Drag:** This type of drag occurs due to the friction between the fluid particles and the surface of the object. The friction is generated by the fluid's viscosity and is responsible for creating a boundary layer on the surface, where the flow slows down due to contact with the solid surface.

In the transonic regime, where the Mach number is close to 1, drag behavior changes significantly due to the formation of shock waves on the surface of the aircraft. This leads to a phenomenon known as drag divergence, where the drag coefficient increases sharply as the flow transitions from subsonic to supersonic speeds. In this regime, compressibility effects play a critical role, and careful consideration must be given to how shock waves and boundary layer interactions influence overall drag (Anderson, 2011).

The drag coefficient (C_D) is a dimensionless measure that reflects the aerodynamic efficiency of a body. It varies with the shape of the body and flow conditions. The total drag coefficient of an aircraft can be expressed by the following relation:

$$C_D = C_{D_0} + kC_L{}^2 \tag{3.11}$$

Where:

- C_{D_0} is the parasite drag coefficient, which includes drag due to surface friction, the shape of the fuselage, landing gear, and wing surfaces;
- k is a constant related to the wing's efficiency and the aircraft's form factor;
- C_L is the lift coefficient, which depends on the wing geometry and flight conditions.

The factor k is closely related to the Oswald efficiency factor (e). The Oswald efficiency factor is a measure of how efficiently the wing generates lift relative to the drag it produces, considering the wing's aspect ratio and the effects of induced drag. Induced drag is the

drag created by the generation of lift, particularly due to the formation of wingtip vortices. The relationship between k and the Oswald factor is given by:

$$k = \frac{1}{\pi e A_R}$$

Where:

- A_R is the aspect ratio of the wing, defined as the ratio of the square of the wingspan to the wing area.

The equation that describes the total drag force on a body is:

$$D = \frac{1}{2} C_D \rho {V_t}^2 S$$

Substituting the expression for C_D, we have:

$$D = \frac{1}{2} \rho {V_t}^2 S \left(C_{D_0} + k {C_L}^2 \right)$$

$$D = \frac{1}{2} \rho {V_t}^2 S C_{D_0} + \frac{1}{2} \rho {V_t}^2 S k {C_L}^2 \tag{3.12}$$

To include the effects of compressibility, the Mach number is used, defined as the ratio of the aircraft's speed to the speed of sound in the medium (a):

$$M_a = \frac{V_t}{a}$$

By considering reference density and pressure as functions of the Mach number, the drag expression can be reformulated. The air density is often related to a reference pressure (P_0) and the adiabatic factor of air γ. To include the effect of the load factor (N_z) and the aircraft's weight, it is necessary to account for the influence of maneuvering, which directly affects induced drag. Therefore, the parametric drag model becomes:

$$\frac{D}{\delta} = \frac{1}{2} P_0 \gamma {M_a}^2 S C_{D_0} + k \cdot \frac{2 {N_z}^2 W^2}{P_0 \delta^2 \gamma {M_a}^2 S} \tag{3.13}$$

This model provides a more comprehensive representation of drag, allowing consideration of both compressibility effects and the influence of additional loads during maneuvers. It is suitable for more varied flight regimes, including transonic and supersonic speeds.

3.2.4 Parameter Estimation Method

Once the pre-processed data are available, the first step is to select the flight segments that will be used for parameter estimation. These segments are carefully chosen to ensure that they represent consistent and stable flight conditions, making them suitable for the analysis.

After selecting the relevant flight segments, the specific energy of the aircraft is calculated based on the derivatives of velocity and altitude variation. Next, the data collected from the onboard acquisition system are transformed into an input variable vector for use in the parametric models developed in the previous sections. The input data vector is represented as follows:

$$X_{data} = \left[\frac{N_r}{\sqrt{\theta}}, W, N_z, M_a, \delta, \Delta\right]$$

These variables are used to estimate the thrust and drag forces using the parametric models. With the estimated values of thrust and drag, a new value for the aircraft's specific energy is calculated. This calculated specific energy is then compared with the actual specific energy derived from the aircraft's flight data.

To adjust the parameters of the models, the least squares method is used. The least squares technique minimizes the sum of the squared differences between the observed and calculated specific energy values. The goal is to find the optimal set of parameters that minimizes the error between these two sets of values. In mathematical terms, the least squares method involves finding the parameter vector θ that minimizes the sum S. The general form of the problem is:

$$S(\theta) = \sum_{i=1}^{n} \left(E_{\text{observed},i} - f(X_{data,i}; \theta)\right)^2$$

Where:

- $f(X_{data,i}; \theta)$ is the model prediction of specific energy at the i-th point, based on the input data $X_{data,i}$ and parameter vector θ.

The method iteratively adjusts the parameter vector θ to minimize $S(\theta)$ by using an optimization algorithm that computes the gradient of S with respect to the parameters:

$$\nabla S = \frac{\partial S}{\partial \theta_1}, \frac{\partial S}{\partial \theta_2}, \ldots, \frac{\partial S}{\partial \theta_m}$$

The optimization algorithm updates the parameters in the direction of the negative

gradient, iterating until the function $S(\theta)$ converges to a minimum, meaning that the calculated and observed values of specific energy are as close as possible.

The algorithm in this case is subject to constraints, as the parameters must remain physically meaningful. These constraints are incorporated into the optimization process, ensuring that the estimated parameters comply with aerodynamic principles and operational limitations.

The optimization process continues until the sum of squared residuals $S(\theta)$ reaches a predefined threshold, or the change in S between iterations becomes negligible. This indicates that the parameters have converged to their optimal values, providing an accurate and reliable representation of the aircraft's performance.

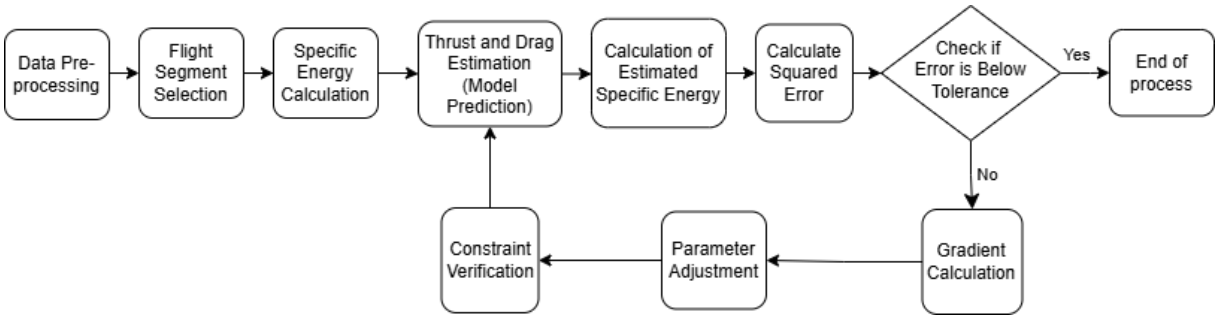


FIGURE 3.2 – Parameter Estimation Process

In this analysis, the parametric models were developed using MATLAB, and the parameter estimation was performed with `fmincon` function, which is well-suited for constrained optimization problems. To validate the results, data from flights not used during the initial parameter estimation phase were employed. This approach aimed to verify the consistency and accuracy of the estimated parameters when applied to independent data. The evaluation of the results and validation of the model's fit were conducted using the R-squared metric, providing a quantitative measure of the match between the observed and predicted values.

4 Results and Discussion

4.1 Evaluation of Climb Time, Distance, and Fuel Consumption

Data from 35 flights conducted during the 2014 flight test campaign in Canoas were made available for analysis. The campaign involved two aircraft, tail numbers 4807 and 4808. These flights took place between August 5 and November 19, under various mission profiles and different meteorological conditions.

The 35 flights were evaluated based on the climb profiles adopted, where the climb speed profiles were verified to determine if they adhered to the performance manual specifications. After this analysis, 14 flights were selected in which the climbs were performed according to the manual, with 9 flights conducted under MIL and 5 under AB conditions.

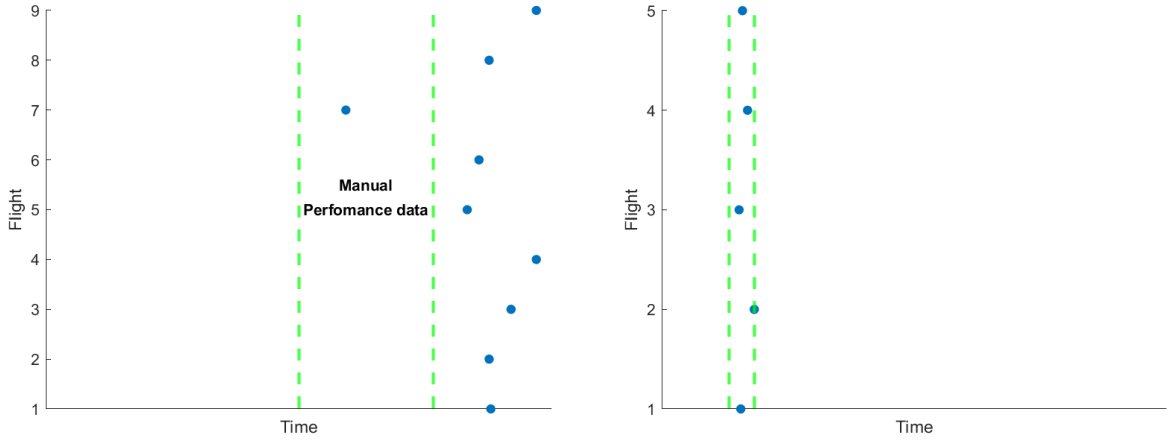
Once these flights were selected, further analysis was performed on the time, distance, and fuel consumption data between 15,000 ft and 25,000 ft. This approach was used to compare the flights under the most similar conditions possible. In addition, the times, distances, and fuel consumption were also calculated for different aircraft weights and atmospheric conditions encountered during the flights using the performance manual.

This evaluation of climb performance focusing on climb time, distance covered, and fuel consumption provides insight into how closely the aircraft performance aligned with the expected values from the performance manual, as well as the influence of varying mission conditions and atmospheric factors on overall climb efficiency.

4.1.1 Time Comparison

Figure 4.1 compares the climb times obtained from the flight data with the estimated times from the performance manual for both MIL and afterburner climb regimes. In the figure, the dots represent the actual climb times recorded between 15000 and 25000 ft, while the green dashed lines indicate the range of performance data from the manual. If the performance manual data and the flight data are in agreement, it is expected that the

climb times of each flight, represented by the blue dots, fall within the region between the green dashed lines.



(a) MIL (b) AB

FIGURE 4.1 – Time comparison.

For the MIL regime, it is clear that the actual climb times are consistently outside and longer than the manual's performance estimates. This discrepancy suggests that, in the MIL regime, the aircraft may not be achieving the performance expected based on the manual, or that the manual's estimates might not accurately reflect the current flight conditions or configurations.

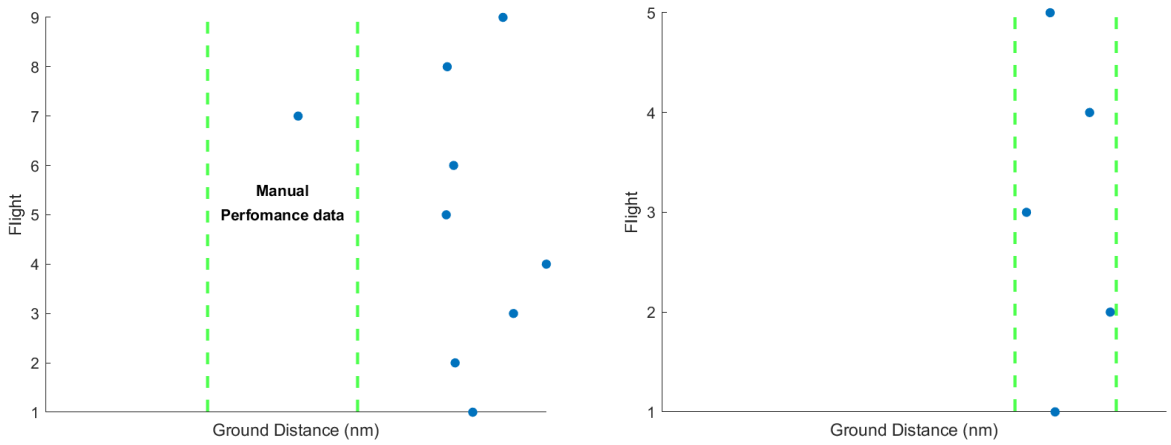
In contrast, for the AB regime, the recorded climb times are more tightly clustered within the range of performance data, falling mostly between the green dashed lines. This indicates that under afterburner thrust, the aircraft's climb times align more closely with the performance estimates, with no significant deviation. The consistency of the climb times with the manual's estimates suggests that the aircraft performs as expected under these conditions.

These observations highlight a noticeable discrepancy in the MIL regime, where the aircraft's actual performance appears to deviate from the performance manual's expectations. On the other hand, in the AB regime, the aircraft's performance is in closer agreement with the manual, indicating a more predictable behavior when operating with afterburner thrust.

4.1.2 Distance Comparison

The analysis of climb distance begins with Figure 4.2, which compares the ground distance obtained by integrating the ground speed with the estimated performance data from the manual, for both the MIL and AB regimes. In the MIL regime, there is a

clear tendency for the performance manual to underestimate the distance required to reach the target altitude compared to the actual flight data. The actual ground distances covered during the MIL climbs are consistently longer than what the manual predicts. This suggests that the aircraft, when operating under military thrust, is climbing less efficiently than expected, reaching the desired altitude with less horizontal displacement.



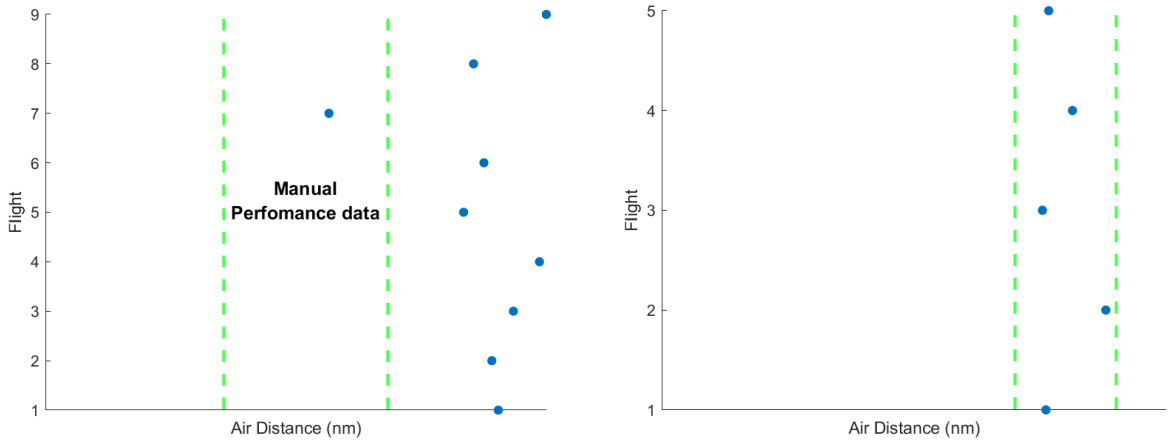
(a) MIL (b) AB

FIGURE 4.2 – Ground distance comparison.

In the afterburner regime, the ground distances align more closely with the performance manual's estimates. The flight data points are within or near the performance range indicated by the green dashed lines, colored performance range, indicating that the climb distances match the manual's predictions more accurately when using afterburner thrust.

It is important to note that ground distance is significantly influenced by wind conditions during the climb. Wind direction and speed can alter the ground distance covered without reflecting the true aerodynamic performance of the aircraft. A headwind can decrease the ground distance, while a tailwind can increase them. Due to this, to remove the wind variable from the analysis, an evaluation based on air distance was conducted.

Figure 4.3 presents the air distance comparison, which eliminates the effect of wind. In the MIL regime the same trend observed in the ground distance analysis persists. The performance manual still underestimates the distance required for the climb, with the actual flight data showing longer air distances than predicted. This reinforces the finding that the aircraft's actual climb performance in military thrust is less efficient than estimated by the manual.



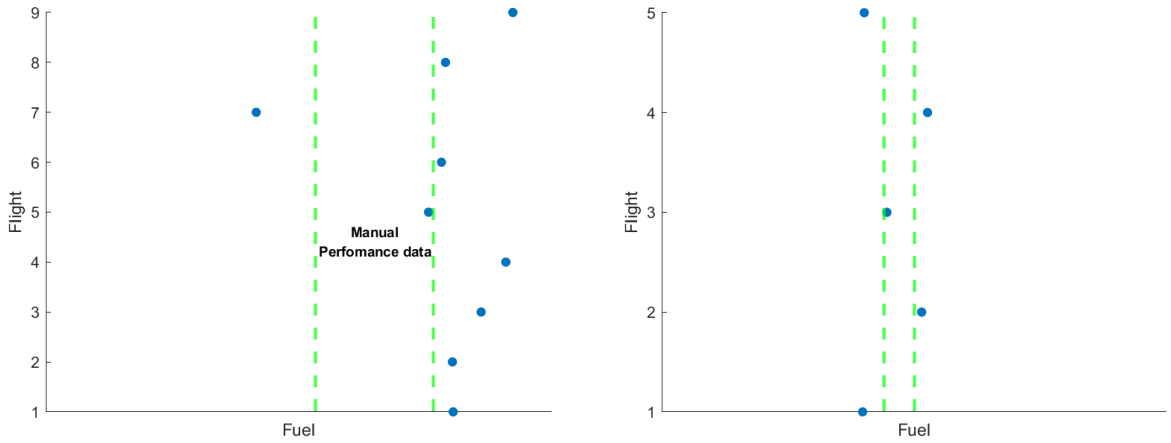
(a) MIL (b) AB

FIGURE 4.3 – Air distance comparison.

For the afterburner regime, the air distances closely match the manual's performance data, similar to the ground distance analysis. The alignment between the flight data and the performance manual in both ground and air distances under afterburner conditions indicates that the manual provides an accurate estimation of the climb distances when afterburner is used, as opposed to the MIL regime where discrepancies remain.

4.1.3 Fuel Comparison

The Figure 4.4 compares the fuel consumption for various flights against the estimated values from the performance manual. In the MIL regime, the fuel consumption for the majority of the flights is clustered near the upper bound of the manual's estimated range. This indicates that the actual fuel consumption is consistently bigger than what the manual predicts, demonstrating a trend of underestimation in the manual's data, though this discrepancy is less pronounced compared to the time and distance parameters discussed earlier.

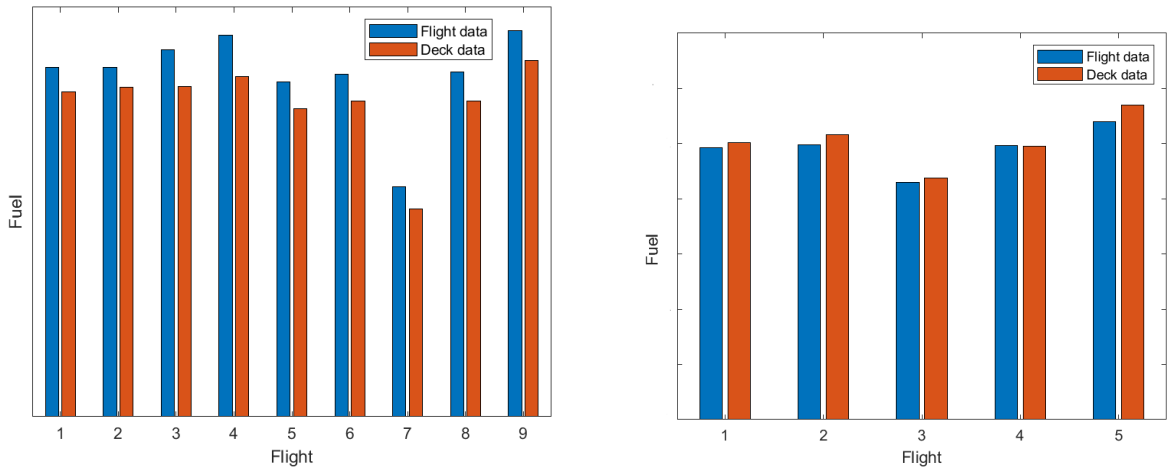


(a) MIL (b) AB

FIGURE 4.4 – Fuel consumption comparison.

For the afterburner regime, the flight data is more evenly distributed across the performance range. There is no clear trend of overestimation or underestimation, which suggests that the performance manual provides a more accurate estimate for fuel consumption when afterburner is engaged. However, there are still some instances of flights consuming slightly more or less fuel than expected.

In Figure 4.5, the fuel consumption estimated by the engine deck is compared to the actual fuel consumption recorded during each flight. The engine deck estimates fuel consumption based on several parameters, including Mach number, altitude, RPM, and the delta temperature relative to ISA, through a lookup in performance tables. The fuel flow from these tables is integrated over the flight duration to calculate the total fuel consumption.



(a) MIL (b) AB

FIGURE 4.5 – Comparison of fuel consumption with engine deck.

In the MIL regime, the fuel flow measured by the aircraft's sensors is consistently

higher than the fuel consumption estimated by the engine deck. This is the same trend compared to the performance manual, where the manual tends to underestimate fuel consumption. In the AB regime, the comparison between the deck data and flight data shows closer alignment, with no significant trend of overestimation or underestimation. The engine deck seems to provide a more accurate estimate for fuel consumption when afterburner is used, consistent with the findings from the performance manual comparison.

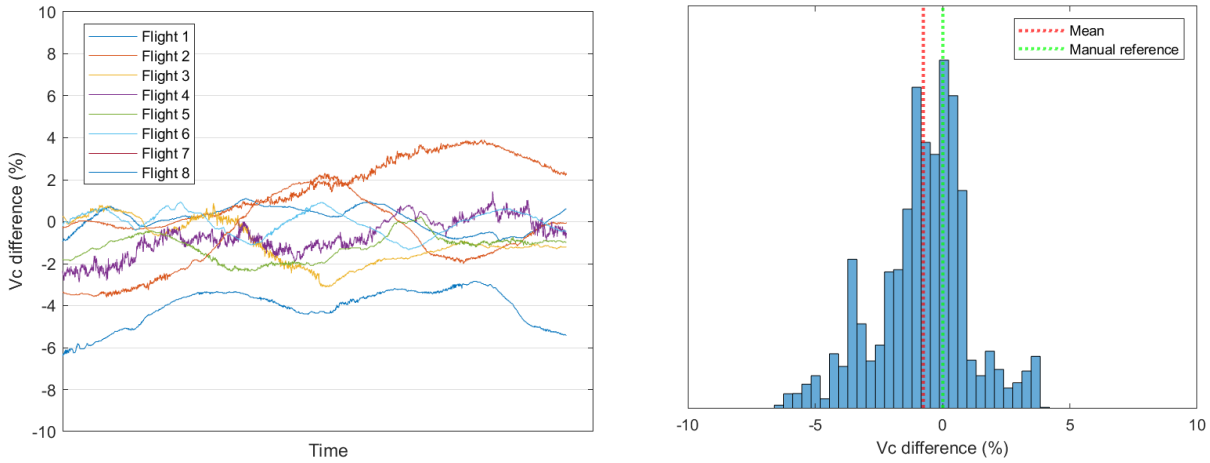
The analysis of fuel consumption across a large number of flights, conducted on different days and using different aircraft, reveals some consistent trends. In the MIL regime, both the performance manual and the engine deck tend to exhibit discrepancies. The performance manual consistently overestimates fuel consumption, while the engine deck data, in contrast, underestimates it compared to the flight sensor data. In the AB regime, both the performance manual and the engine deck offer more accurate estimates, with no clear trend of over- or underestimation.

The results indicate that the aircraft's performance in military thrust is less efficient than predicted by the manual. However, it is important to highlight that other sources of variability, such as adherence to the recommended climb profiles, may also influence these results. This will be explored further in the next section.

4.2 Velocity Profile Analysis

The objective of this section is to evaluate how closely the pilot adhered to the recommended climb profile specified in the performance manual. Adhering to the correct climb speed is crucial, as deviations from the optimal speed can significantly affect key performance parameters such as climb time, distance, and fuel consumption. If the aircraft is flown at speeds higher or lower than the ideal, the performance can be negatively impacted, potentially leading to discrepancies between the observed and expected performance metrics.

Figure 4.6 illustrates the climb speeds for each flight in the MIL regime relative to the speed defined by performance manual. In this analysis, calibrated airspeed (V_c) was used for comparison. V_c is the indicated airspeed corrected for instrument and position errors, providing a more accurate measurement of the actual airspeed under standard atmospheric conditions. In Figure 4.6a, the speed profiles for the individual flights are shown over time, with each line representing a different flight. In an ideal scenario, all flights would closely follow the recommended climb speed from the manual, minimizing performance deviations.

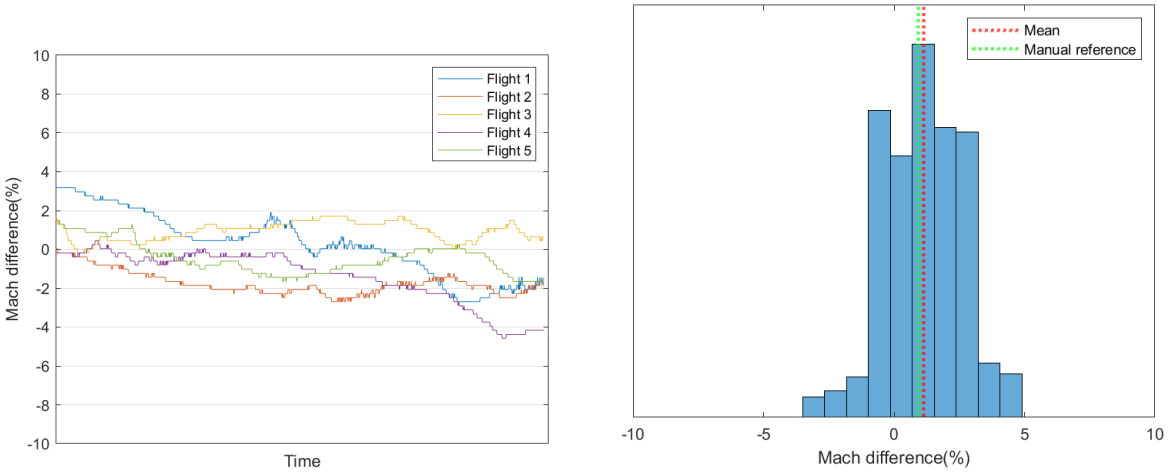


(a) (b)

FIGURE 4.6 – Evaluation of the speed profile for the MIL regime.

Figure 4.6b presents a histogram of the climb speeds, showing that the average speed across all flights is slightly lower than the recommended speed from the manufacturer, by approximately 1.35%. This marginal difference indicates that the pilots adhered quite closely to the recommended climb profile, and thus this slight speed variation is unlikely to be a significant factor contributing to the discrepancies observed.

A similar analysis was conducted for the afterburner regime. In this case, Mach number was used as the reference parameter, as it is the primary factor specified by the performance manual for climbs in this regime. As with the MIL regime, the climb speeds were compared with the manual's recommendations to assess any deviations.



(a) (b)

FIGURE 4.7 – Evaluation of the speed profile for the AB regime.

The results of this analysis were similar to those of the MIL regime, with the average Mach number being 1.23% higher than the value recommended by the performance

manual. This slight deviation suggests that the pilots followed the recommended climb profile closely, and this small difference is unlikely to be the primary cause of any observed discrepancies in performance metrics such as climb time or fuel consumption.

Given this analysis, it can be concluded that the velocity profile during the climbs was not a major contributing factor to the overestimation trends identified in the performance manual, particularly for the MIL regime. Further investigation into other factors, such as engine performance, is likely required to fully explain the discrepancies observed.

4.3 Thrust Analysis

In this section, thrust was calculated using an estimation methodology, as described in subsection 3.2.4, to compare the estimated thrust values with those provided by the engine deck data. The objective was to identify any performance discrepancies that might help explain the observed differences between the experimental flight data and the manual's predictions. By analyzing these discrepancies, we aim to better understand potential factors impacting engine performance in real flight conditions.

4.3.1 Problem constraints

In the process of estimating thrust using parametric models, several constraints were applied to ensure that the parameters remained physically meaningful. These constraints were essential to maintain the reliability and accuracy of the results.

The first constraint applied was ensuring that the coefficient C_0 in the parametric thrust model was positive. This guarantees that the model will predict a positive thrust value, which is necessary for the physical feasibility of the model.

Further constraints were applied to the drag model parameters, particularly regarding the parasite drag coefficient C_{D_0} and the factor k, which is directly related to the Oswald efficiency factor. For aircraft similar to the F-5, the Oswald efficiency factor (e) typically ranges between 0.7 and 0.9 (Samoylovitch; Strelets, 2000). This provides a corresponding range for the factor k to fall between 0.09 and 0.13.

The parasite drag coefficient C_{D_0} also required careful consideration. For the F-5, C_{D_0} typically ranges between 0.015 and 0.03. However, due to the flight configuration involving external fuel tanks, which increase drag, a wider range of 0.015 to 0.05 was considered in the model.

By applying these constraints, the parametric models are better aligned with the physical behavior of the aircraft, ensuring that the results are both realistic and meaningful.

4.3.2 Data Selection

For the parameter estimation process, data were selected from specific segments of one of the flights in the campaign where there were no abrupt variations in the nozzle area, engine rotational speed, or $\dot{H}/V$. As shown in Figure 4.8, the selection of these stable segments allowed us to explore a wide range of input variables, including N_r, Mach number, and Δ.

This careful selection of flight segments ensured that the data used for parameter estimation were both consistent and representative of various operating conditions, which is crucial for building reliable parametric models.

To validate the models developed from these segments, two flights were selected for further analysis. One flight involved a climb in the MIL thrust regime, while the other was conducted in the afterburner regime. By comparing the model predictions to the actual flight data in these different operating regimes, it is possible to assess the accuracy and generalizability of the estimated parameters across a range of flight conditions.

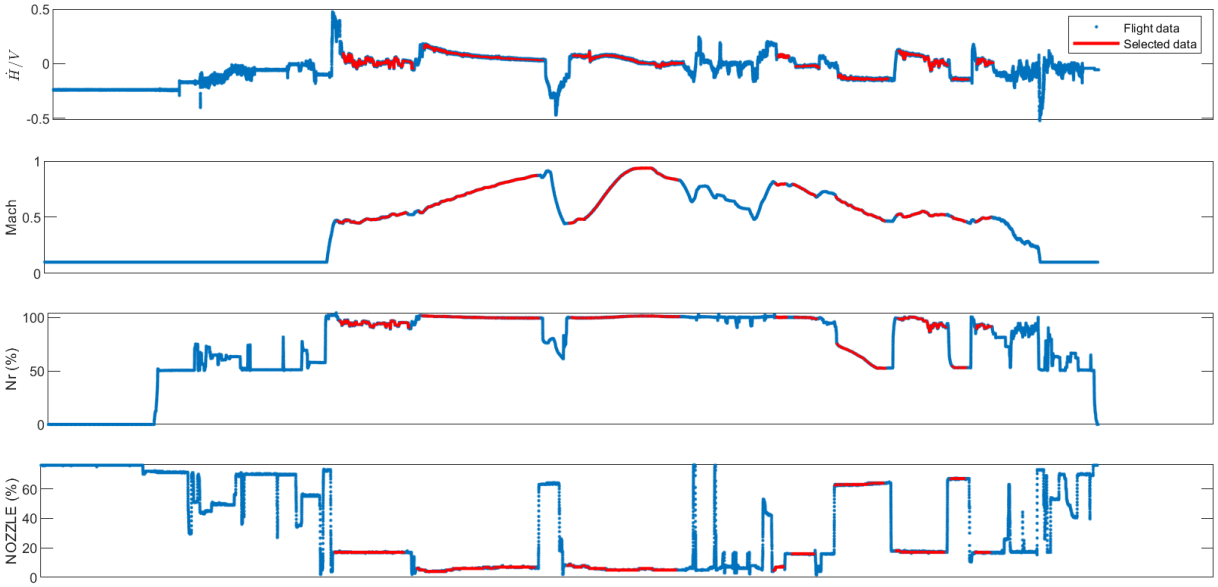


FIGURE 4.8 – Selected data for parameter estimation.

4.3.3 Results of parameter Estimation and Validation

The results of the parameter estimation process are shown in Figure 4.9, where the fit of the model to the selected data is presented. The model validation is illustrated in Figure 4.10, where the R-squared values for the MIL and AB regimes were 0.89 and 0.86, respectively. While these values indicate a reasonably good fit, the model was not able to accurately capture the dynamics of the final phase of the flight involving approach and landing. These phases of flight, characterized by rapid changes in power settings and

aircraft configuration, likely contributed to the relatively lower adherence to the flight data observed.

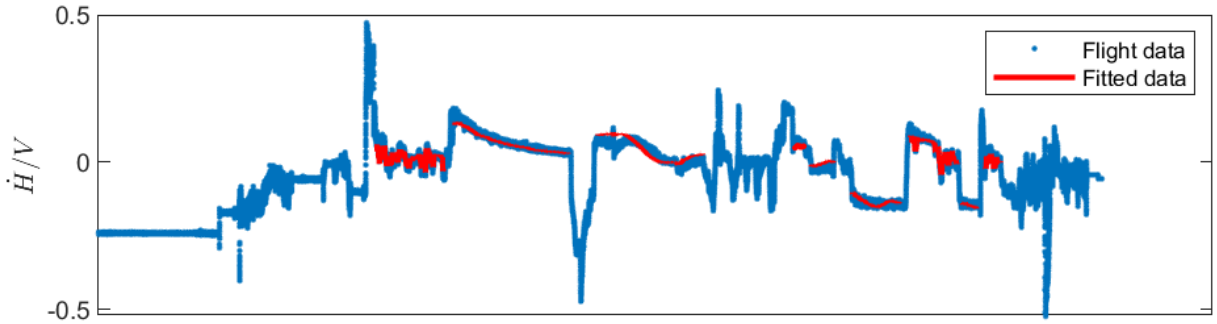
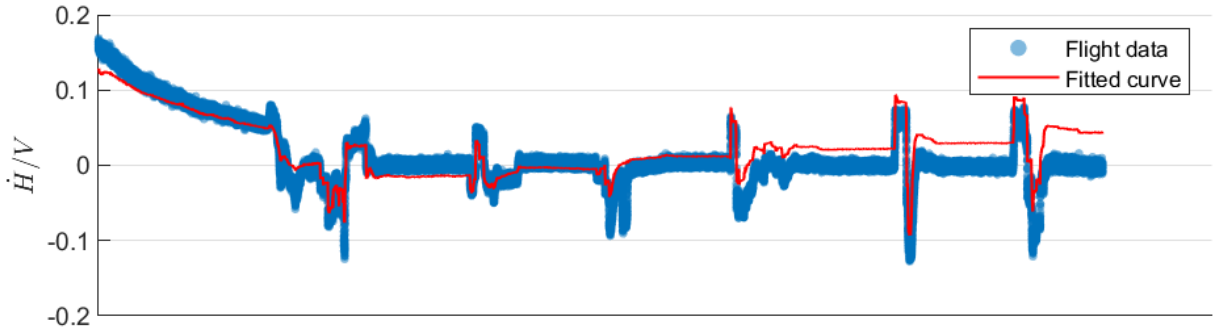
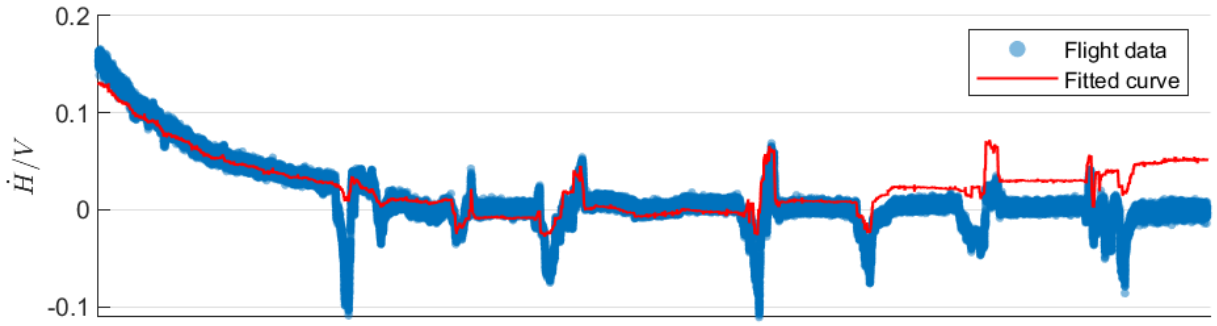


FIGURE 4.9 – Result of parameter estimation.

Despite the challenges in modeling these phases, the propulsion model demonstrated significantly better accuracy during the cruise climb segment, which is the primary focus of this study. In this phase, the model achieved R-squared values of 0.98 and 0.99 for the MIL and AB regimes, respectively, indicating very good agreement with the flight data during this portion of the flight. This suggests that while the model may have limitations in capturing the full flight envelope, it performs well in the segments most relevant to this analysis.



(a) MIL climb

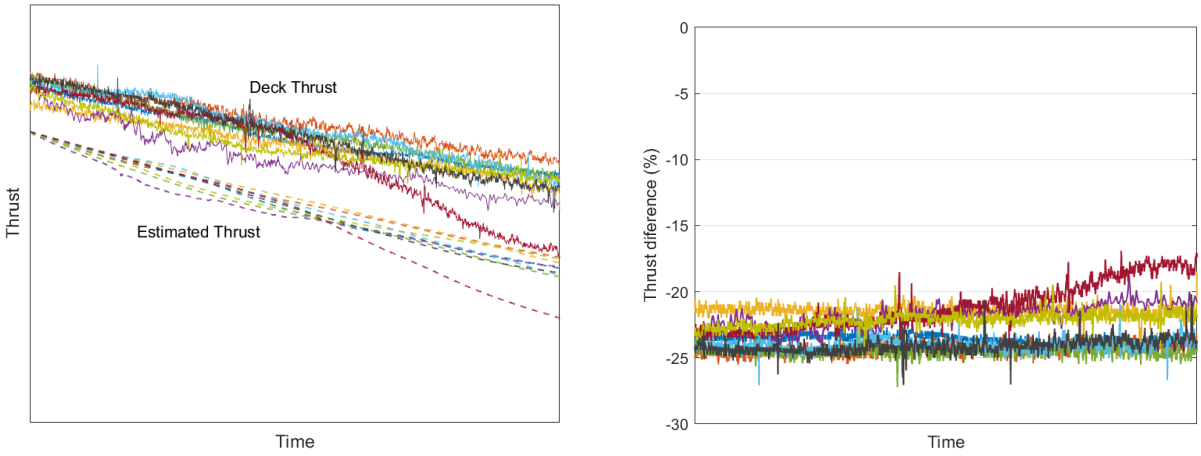


(b) AB climb

FIGURE 4.10 – Validation of the estimated model.

4.3.4 Estimated Thrust

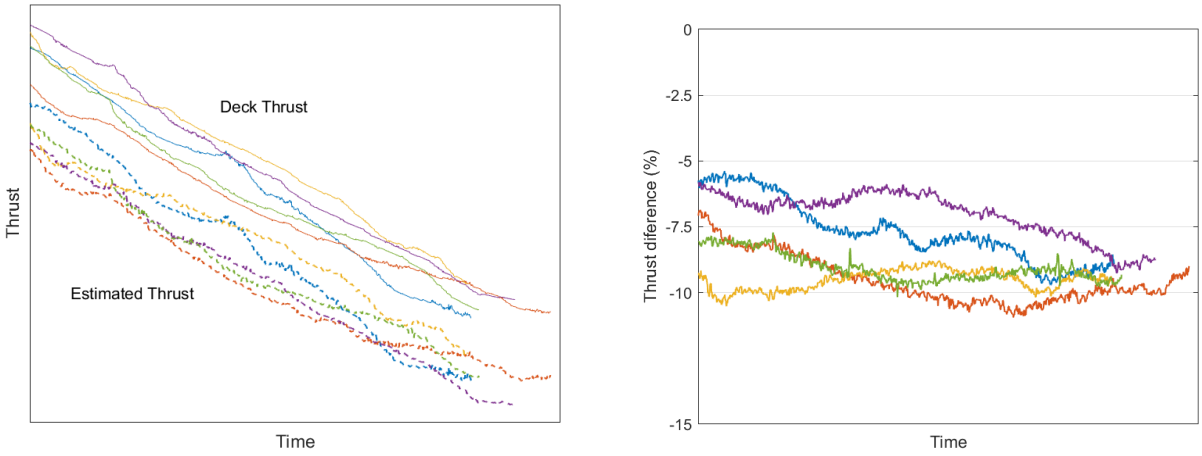
The next two figures compares the thrust values obtained from the proposed propulsion model with those derived from the engine deck, which uses parameters such as N_r, Mach number, altitude, and nozzle opening. In the figures, the colored lines represent the thrust values for each flight, with the solid lines showing the data from the engine deck and the dashed lines representing the values from the proposed model.



(a) (b)

FIGURE 4.11 – Thrust during the climb in MIL regime.

As seen in both figures, the model correctly captures the same trend of decreasing thrust during the climb, corresponding to the decrease in atmospheric pressure as altitude increases. For the MIL regime (Figure 4.11), the estimated thrust values were 22% lower than the values provided by the engine deck. On the other hand, for the AB regime (Figure 4.12), the estimated thrust was underestimated by 8% compared to the thrust provided by the engine deck.



(a) (b)

FIGURE 4.12 – Thrust during the climb in AB regime.

The thrust values provided by the engine deck consider an uninstalled engine with 100% inlet efficiency, without accounting for compressor bleed, power extraction, installation drag, or the aerodynamic effects present when the engine is mounted on the airframe. This makes it necessary to distinguish between gross thrust, measured in a controlled test environment, and net thrust, which is the effective thrust available during actual flight operations.

Gross thrust is the total thrust produced by the engine in isolation, typically measured in a test bench environment where there are no aerodynamic interactions with the aircraft structure. Net thrust, in contrast, is the thrust available to propel the aircraft during flight, taking into account installation-related drag forces that reduce the effective thrust.

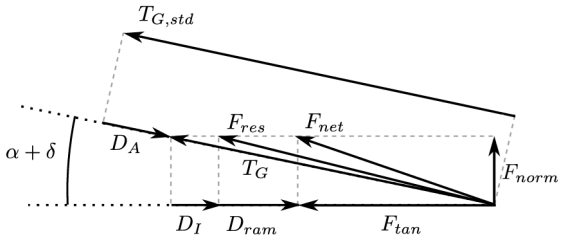


FIGURE 4.13 – Decomposition of forces acting on the propulsion system (Vitsas, 2015).

As illustrated in Figure 4.13, net thrust F_{net} is a function of standard gross thrust TG_{std}, the interference drag from the exhaust nozzle D_A, ram drag D_{ram}, inlet drag D_I, and the angles of attack α and the fixed incidence between the exhaust axis and the longitudinal axis of the aircraft, represented by δ.

It is important to note that the methodology described in section 3.2.2 is not capable of determining the gross thrust T_G. This limitation is due to the challenge of accurately quantifying the inlet drag D_I and exhaust drag D_A and isolating these values from the resulting net thrust. Consequently, the thrust models derived from flight test analysis are capable of accurately calculating the net thrust T_n.

Several sources of drag impact net thrust when the engine is integrated with the aircraft:

- **Inlet Drag**: The engine inlet generates drag as it directs incoming air into the engine. This drag includes both the frictional resistance along the inlet surfaces and the pressure losses that occur as the air slows down and is compressed before entering the compressor.

- **Ram Drag**: Ram drag results from the momentum of the air entering the engine. As the aircraft speed increases, the airflow entering the engine exerts an opposing force, known as ram drag, which effectively reduces the available net thrust, particularly at high speeds.

- **Nozzle Internal Drag**: Internal drag arises as exhaust gases pass through the engine's nozzle. This resistance results from friction and pressure losses within the nozzle duct, especially during high-flow operations, such as afterburner, where exhaust flow velocity is significantly increased. This nozzle drag reduces the effective net thrust by dissipating part of the energy in the exhaust flow.

Research by Stańkowski et al. (2016) and Zhang et al. (2015) emphasizes the importance of these installation drag sources in assessing thrust performance in flight. Stańkowski et al. describe the influence of pylon-wing interactions and nacelle aerodynamics on net thrust, while Zhang et al. report that internal nozzle drag can account for 15-20 drag counts, significantly affecting the net thrust observed in flight (Stańkowski; MacManus; Sheaf; Christie, 2016; Zhang *et al.*, 2015).

While a difference between the thrust measured by the parametric model and the engine deck data is expected, a discrepancy of 22% in the MIL regime, compared to an 8% reduction in the AB regime, suggests that factors beyond unaccounted installation drag may be influencing MIL performance. This larger difference in the MIL regime could indicate potential efficiency losses or that the engine's operating point in-flight may not perfectly align with the simulated operating conditions assumed in the deck data. To reach a more accurate diagnosis, further in-depth investigation is required.

Figure 4.14 compares the EGT values recorded by the aircraft's onboard sensors with the data provided by the engine deck for each of the MIL climbs. The figure clearly shows that, for all flights, the engine operates at a lower EGT than the values predicted by the deck. This consistent difference suggests that the engine is operating cooler than expected during MIL climbs.

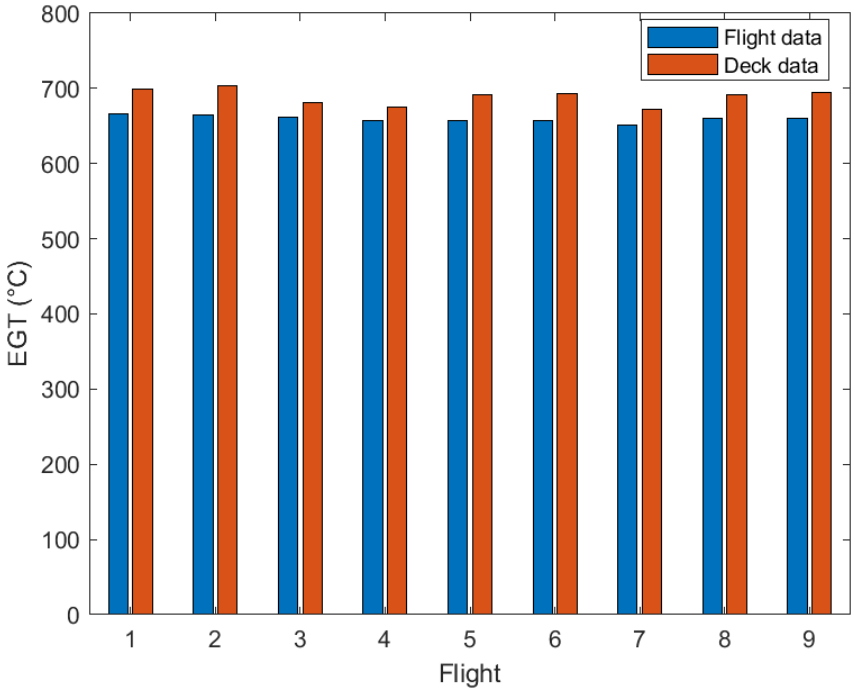


FIGURE 4.14 – EGT comparison in MIL regime.

The relationship between thrust and the combustion chamber temperature is well established, as a higher combustion temperature typically correlates with increased thrust production. However, directly measuring the combustion chamber temperature is challenging due to its extreme conditions. EGT, which measures the temperature of the exhaust gases leaving the engine, serves as an indirect indicator of combustion chamber temperature and overall engine performance (Saravanamuttoo; Rogers; Cohen, 2001). Lower EGT values, as seen in Figure 4.14, can indicate lower energy output from the engine, which could help explain the reduced thrust observed in the MIL regime.

The consistently lower EGT measured during the MIL regime corroborates the thrust findings, suggesting that the engine is not producing as much power as expected based on the engine deck data. This reduced EGT aligns with the 22% underestimation of thrust found in the parametric model for the MIL regime, indicating potential efficiency losses or operational deviations from the expected deck performance.

Unfortunately, the EGT data provided by the engine deck for the AB regime was inconsistent, making it impossible to perform a reliable comparison for that regime. It is important to note that the statements regarding the engine's performance are hypotheses based on the analysis of the available data and the results of a relatively simplistic thrust model. To definitively conclude whether the engine is indeed underperforming or operating outside the expected conditions, a more in-depth evaluation is required.

5 Conclusions

This study identified discrepancies in the estimated climb performance data for the F-5 aircraft's J85-GE-21 engine in the MIL regime when compared to experimental flight data. The analysis revealed that the actual time, distance, and fuel consumption during climbs consistently exceeded the predictions provided by the performance manual, suggesting potential limitations in the manual's to estimate performance parameters or efficiency losses in real-flight conditions. These discrepancies were more pronounced in the MIL regime, while in the AB regime, the performance manual's data aligned more closely with observed results, indicating more accurate estimations under afterburner conditions.

The parametric modeling approach used in this work is relatively simple and, therefore, not fully capable of capturing the complex dynamics of the engine's operation, including the interactions between its various systems and components. While this model was effective in identifying general thrust trends during the cruise climb phase, it demonstrated limitations in accurately representing the engine's behavior across all flight phases, especially in scenarios requiring rapid power adjustments, such as approach and landing. A significant limitation of this methodology is its reliance on a static atmospheric condition, which does not account for the effects of wind.

A significant challenge in this study was that the engine deck data available for reference considers an uninstalled engine, without accounting for compressor bleed and power extraction. This discrepancy between the engine deck's gross thrust values and the net thrust estimated by the parametric model complicated direct comparison, especially given the lack of specific studies in the literature detailing expected percentage differences between gross and net thrust. This gap made it challenging to contextualize the observed discrepancies within an established framework.

Future research should focus on adopting advanced modeling software capable of providing a more accurate and comprehensive simulation of engine behavior, which would capture these operational nuances more effectively. Additionally, it would be beneficial to analyze the impact of wind on climb performance parameters, as varying wind conditions could influence the observed discrepancies. Expanding the dataset to include a larger number of flights under varied conditions would further strengthen the analysis, en-

abling a deeper understanding of the discrepancies observed and contributing to improved performance estimates and overall flight safety.

References

ANDERSON, J. **EBOOK: Fundamentals of Aerodynamics (SI units)**. [*S.l.*]: McGraw hill, 2011. Cit. on p. 32.

DOMINGOS, F. A. **Estimativa Simultânea da Curva de Tração de um Motor Turbofan e da Polar de Arrasto a partir de Ensaios em Voo**. 2023. Graduation Thesis – Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, Brazil. Cit. on p. 31.

EMBRAER. **Flight Manual: Aircraft F-5EM (Single-Place) F-5FM (Two-Place)**. Revision 6. São José dos Campos, Brazil, Nov. 2010. Cit. on pp. 19–21.

FAB. **Força Aérea Brasileira - Dimensão 22**. Accessed: 2024-06-20. 2024. Available from: `https://www.fab.mil.br/aeronave#22`. Cit. on pp. 18, 19.

MCCLAMROCH, N. H. Steady Aircraft Flight and Aircraft Performance. **Ann Arbor**, Citeseer, v. 1001, S 48, p. 109–2140, 2006. Cit. on p. 28.

ROSENBERG, R. E.; SCHUCHT, G. The MCA Method of Determining Thrust of Jet Aircraft in Flight. **Journal of Aircraft**, American Institute of Aeronautics and Astronautics, v. 22, n. 10, p. 885–889, 1985. DOI: `10.2514/3.45220`. Cit. on pp. 23, 28–30.

SAMOYLOVITCH, O.; STRELETS, D. Determination of the Oswald efficiency factor at the aeroplane design preliminary stage. **Aircraft Design**, Elsevier, v. 3, n. 3, p. 167–174, 2000. Cit. on p. 43.

SARAVANAMUTTOO, H. I.; ROGERS, G. F. C.; COHEN, H. **Gas turbine theory**. [*S.l.*]: Pearson education, 2001. Cit. on pp. 30, 49.

SCHUCH, G.; WEISS, P.; KRAUSE, P.; THURLING, A. J. Determination of Aircraft Performance without an Available Engine Thrust Deck: Applying EFMA (Excess Force Mass Acceleration) Flight Test Technique and Evaluation Method. **Proceedings of the USAF Test Pilot School and German Armed Forces Technical and Airworthiness Centre for Aircraft**, Edwards Air Force Base, USA, 2001. Available from: `https://doi.org/10.2514/3.45220`. Cit. on p. 23.

STAŃKOWSKI, T. P.; MACMANUS, D. G.; SHEAF, C. T.; CHRISTIE, R. Aerodynamics of aero-engine installation. **Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering**, Sage Publications Sage UK: London, England, v. 230, n. 14, p. 2673–2692, 2016. Cit. on p. 48.

USAF. **Performance Phase. Chapter 11. Cruise Performance Theory**. Edwards AFB, CA, 1993. 9-31 p. (Series, ADA320213). Available from: https://apps.dtic.mil/sti/citations/ADA320213. Visited on: 15 Oct. 2024. Cit. on p. 30.

VITSAS, P. A. A Theoretical Review of the EFMA Flight Test Method. **46th SFTE International Symposium, Lancaster, CA**, International Test Pilot School, London, Ontario, Canada, 2015. Available from: https://doi.org/10.2514/3.45220. Cit. on pp. 23, 28, 30, 47.

ZHANG, Y.; CHEN, H.; FU, S.; ZHANG, M.; ZHANG, M. Drag prediction method of powered-on civil aircraft based on thrust drag bookkeeping. **Chinese Journal of Aeronautics**, Elsevier, v. 28, n. 4, p. 1023–1033, 2015. Cit. on p. 48.

FOLHA DE REGISTRO DO DOCUMENTO

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TC	31 de outubro de 2024	DCTA/ITA/TC-056/2024	53

5. TÍTULO E SUBTÍTULO:

Evaluating J85 engine thrust chart accuracy from flight test data

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11. RESUMO:

The accuracy of in-flight performance estimates is crucial for effective mission planning and operational safety of the F-5 aircraft. Observations from experimental flights indicated discrepancies in the climb data for the J85-GE-21 engine, especially in the MIL regime, prompting a detailed investigation into these differences. This study aims to assess the accuracy of the performance manual's predictions by comparing them with data collected from real flights. The analysis focused on two operational regimes: MIL and AB. In the MIL regime, it was found that the actual climb times, fuel consumption, and distances consistently exceeded the estimated values provided in the manual. In the AB regime, however, the experimental data showed less discrepancy with the manual's estimates, indicating higher accuracy. The speed profiles were analyzed, and the net thrust was estimated using the Mass Consumption Acceleration (MCA) method, based on the conservation of specific energy. The parametric model developed to represent the thrust demonstrated high accuracy, achieving a coefficient of determination of 0.98 in the MIL regime and 0.95 in the AB regime, providing reliable estimates for both regimes during climbs. Comparison with the gross thrust data from the engine deck revealed an average difference of 22% in the MIL regime and 8% in the AB regime. Although differences between gross and net thrust are expected, the magnitude of the discrepancy in the MIL regime suggests that the engine operates below the expected performance level.

12. GRAU DE SIGILO:

(**X**) **OSTENSIVO** ( ) **RESERVADO** ( ) **SECRETO**